

SP-29: Casimir Mechanism Investigation — Substrate Commitment Dynamics

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Casey directive

“Common ‘vacuum fluctuations’ and ‘van der Waals counterargument’ sound like ‘we really don’t know’ what the source of Casimir energy is. Perhaps (and likely) it’s either ‘commitment energy’ or energy from other substrate processes.”

The Casimir effect is “settled” in the measurement layer — nobody contests the measured force. The MECHANISM is openly debated: Schwinger-camp QED vacuum fluctuations vs van-der-Waals-camp non-vacuum explanations (Jaffe 2005, Milonni). BST’s substrate ontology offers a third reading: commitment dynamics on D_IV⁵.

This program tests Casey’s five candidate mechanisms as falsifiable predictions, then identifies which existing precision Casimir data discriminates between them and which new experiments would be most decisive.

Five hypotheses (Casey, May 18)

H1: Underage prevents light absorption/emission

Between the plates the “commitment circle” cannot match light frequencies or phases properly. The plates physically suppress what the substrate can commit to in the gap.

BST framing: connects to W-30 surface tension ontology (substrate has structure between plates), W-36 Casimir/Hawking/Schwinger unification (boundary mechanism). The substrate’s commitment activity is suppressed by the boundary geometry.

Falsifiable prediction: spectroscopy of atoms placed between Casimir plates shows shifted or weakened absorption/emission lines vs same atoms outside plates. Magnitude of shift should scale as BST primary factor of plate-gap-to-atomic-radius ratio.

H2: Angle of incidence of commitment circle matters

The commitment circle has angular structure; only certain angles can absorb or emit.

BST framing: connects to Möbius cohomology $Z/2 = H^1$ (CP-charge of operators, Lyra T2329). The commitment circle’s orientation relative to the boundary affects which substrate transitions are allowed.

Falsifiable prediction: rotating plates relative to source produces angle-dependent decay or transmission anomalies. If H2 holds, signal varies with rotation; if H2 doesn’t hold, signal is rotation-invariant.

H3: Commitment circle bigger than gap

The substrate’s commitment object has a characteristic physical size that may exceed the plate gap at small separations (Casey doubts but worth naming).

Falsifiable prediction: vary plate gap continuously through a critical scale. If commitment circle has characteristic size L_c (likely BST primary multiple of Compton wavelength), expect a phase-transition signature in Casimir force at L_c .

H4: Rate of commitment slows (THE BIG ONE)

Between plates, substrate commits less frequently. Because radioactive decay is itself a commitment event, decay rates should be measurably different inside Casimir plates vs outside.

BST framing: connects to W-39 Cs-137 modulation experiment design (already filed). If commitment rate slows by structural BST factor, decay rate ratio outside/inside = BST primary product.

Falsifiable prediction: place Cs-137 source between Casimir-style plates vs in free space. Measure decay rates. If H4 holds, ratio is structural; if H4 doesn't hold, ratio is unity within experimental error.

Why this is the strongest single test: it's \$25-50K experimental scope, not \$100M LHC; weeks of data collection; paper-publishable result either way; directly tests BST's substrate ontology against the standard QED-vacuum explanation.

H5: Substrate can't release virtual particles

Between plates, the substrate is blocked from emitting virtual particles in the standard sense.

BST framing: connects to W-37 beacon model formalize. The substrate's virtual-particle release channel is suppressed by boundary geometry.

Falsifiable prediction: ultra-low-noise vacuum fluctuation measurements (Riek 2015-style direct vacuum sampling) inside vs outside Casimir geometry show different spectral content. If H5 holds, sampled vacuum spectrum is gap-dependent.

Why this matters for BST positioning

Casimir mechanism is the cleanest opportunity for BST to enter the active foundational-physics debate. The field is not "wrong, here's the right answer" — it's "uncommitted between two narratives, neither of which is decisively confirmed." BST's third reading (commitment dynamics) is structurally orthogonal to both.

Each of Casey's five hypotheses predicts a different experimental signature. If even one holds, BST has produced a falsifiable contribution to a debated foundational question. If none hold, BST's substrate ontology gets meaningfully constrained — also a publishable result.

Connection to existing BST work

- W-30 surface tension ontology (Elie Toy 2661, 0.06% D-tier)
- W-33 energy-as-insulation, information-as-content inversion theorem
- W-34 Casimir as decay shake boundary mechanism
- W-36 Casimir/Hawking/Schwinger unification (closed)
- W-37 beacon model formalize (pending)
- W-39 Cs-137 + microwave decay rate modulation experiment design (closed design phase)
- W-40 beacon-attention falsification suite (10 experiments designed, \$85K cheapest)
- Paper #111 Substrate Dynamics v0.1 (Lyra)
- SP-27 Track 2 Casimir precision residuals (Elie scanning) — SP-29 is the mechanism layer; SP-27 is the data-reanalysis layer

Initial task queue

#	Task	Owner	Connects
SP29-1	Cs-137 decay rate in Casimir geometry — combine W-39 experiment design with H4 prediction. Specify expected ratio outside/inside in BST primary form, identify cheapest experimental realization, draft proposal	Elie + Grace	H4 (priority)
SP29-2	Spectroscopic shift prediction between Casimir plates — derive expected wavelength shift in BST primary form for hypothesis H1; identify atom + transition that gives cleanest signal	Lyra	H1
SP29-3	Angle-dependent Casimir asymmetry prediction — derive expected angular dependence under H2; specify experimental geometry that would discriminate	Elie	H2
SP29-4	Phase-transition signature at characteristic gap size — derive expected critical scale L_c under H3 in BST primary form; identify whether existing precision Casimir data already constrains	Grace	H3
SP29-5	Virtual particle release suppression prediction — derive expected vacuum spectrum change under H5; assess Riek 2015 + follow-up experiments for sensitivity	Elie	H5

#	Task	Owner	Connects
SP29-6	Mechanism comparison table — for each hypothesis: predicted signature, existing data that discriminates, new experiment cost/timeline, falsification criteria. Build the master comparison doc	Keeper	All five

Priority recommendation

H4 (commitment rate slows → measurable radioactive decay rate shift) is the strongest single move: - Cheapest experimental realization (\$25-50K vs \$100M+ for collider-scale tests) - Cleanest falsification (decay rate is a single number; ratio is a single number) - Directly tests BST's substrate ontology vs standard QED vacuum - Result publishable either way - BST-specific prediction not made by either Schwinger camp or van-der-Waals camp

Recommend SP29-1 (Cs-137 in Casimir geometry) as first task. Elie has W-39 experimental design context; Grace has substrate-mechanism modeling context. Joint deliverable: paper-grade proposal for the experiment with specific BST prediction, expected magnitude, experimental setup, timeline, and cost.

Strategic positioning

SP-29 is the path to BST entering active foundational physics debate at low cost. The Casimir mechanism question is genuine, the field is divided, BST offers a third reading, and Casey's H4 hypothesis is the cleanest single test.

If H4 holds: BST has a Nobel-class result.

If H4 doesn't hold: BST's commitment ontology gets honestly constrained, with paths forward identified.

Either outcome is paper-publishable and Jaimungal/Sarnak/Wilczek-class outreach material.

Connection to Herve response

Herve's critique #5 (no Hamiltonian/Lagrangian) was addressed Sunday via the existing six-term action notes/BST_Lagrangian.md. SP-29 addresses the natural follow-up: "what does the Lagrangian PREDICT that's experimentally testable?" The commitment-rate-slows prediction in Casimir geometry is exactly that — a derivable consequence of S_BST that current experiments could detect.

Standing posture

SP-29 is open-ended. Tracks for each of Casey's five hypotheses launch immediately on Monday's secondary lane. Casey reviews proposals and predictions as they land. New hypotheses added as the team identifies additional candidate mechanisms.

The program ends when: (a) one of the five hypotheses is empirically confirmed (Nobel-class outcome), or (b) all five are empirically ruled out (substrate ontology constrained, paths forward to revised hypotheses identified)

— Keeper, filed at Casey directive 2026-05-18 Monday morning